Robot Model Evaluation Results

FIF, CFMPE, FMPE, FFF and M-FFF results for FR3 5DOF and 6DOF, with separate appendices for Umut's planar kinematics and ballistics evaluations.

6DOF Validation Results

200 validation poses × 1000 model samples per pose. Err_post and Err_resim are arithmetic means. Rows are sorted by increasing Err_post within this table only.

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	FMPE Midpoint 32	0.02052	0.00323	5.83764 / 16.95654	1.42017 / 4.17553	0.07935	0.96259	75.86588*
2	FMPE Euler 64	0.02060	0.00297	4.47107 / 13.71121	1.07971 / 3.63905	0.07491	0.96424	75.87196*
3	FMPE Midpoint 16	0.02067	0.00307	6.26979 / 16.39942	1.37532 / 3.73115	0.08168	0.96428	37.86991*
4	FMPE Euler 32	0.02104	0.00417	4.88055 / 14.07129	1.08067 / 3.47117	0.07150	0.96253	37.92870*
5	FMPE Midpoint 8	0.02125	0.00533	8.51894 / 23.12528	1.77234 / 4.57218	0.08732	0.96068	14.12846*
6	CFMPE Midpoint 32	0.02580	0.00613	7.05520 / 26.09504	1.58730 / 5.91851	0.08879	0.92179	61.79415
7	CFMPE Euler 64	0.02597	0.00573	5.58481 / 22.86129	1.25875 / 5.42524	0.08439	0.92590	61.89709
8	CFMPE Midpoint 16	0.02597	0.00569	6.91483 / 23.52885	1.45728 / 5.49150	0.09103	0.92485	30.62228
9	CFMPE Midpoint 8	0.02633	0.00676	8.67135 / 26.18492	1.71005 / 5.53078	0.09468	0.92522	15.00537
10	CFMPE Euler 32	0.02640	0.00624	5.57651 / 21.47007	1.18817 / 5.12935	0.08095	0.92665	30.87547
11	M-FFF torus wrapped	0.11292	2.17862	105.20305 / 277.01604	61.86363 / 166.28867	0.29480	0.20574	0.18355
12	FFF y noise 0.01000	0.12776	3.93056	36.51700 / 206.37500	95.10297 / 171.42883	0.23541	0.02332	0.17911
13	FFF Euclidean	0.13003	3.98663	48.42435 / 273.51988	99.91498 / 172.07097	0.29622	0.02756	0.17673
14	FIF	0.14995	1.90710	184.54767 / 852.33822	43.34425 / 161.15251	0.06663	0.00147	0.61916

Local FIF/CFMPE: GTX 1650 Ti. Emine: RTX 3060 Laptop. All FR3 training seeds are 0. Time measures model sampling only; timings on different hardware do not support a direct speed ranking.

Reference note: the local validation cache hashes differ from those reported by Emine. The target poses and sample counts are shared, but the reference samples are not identical.

Results are rounded to five decimal places. A displayed 0.00000 can represent a small nonzero value.

* Emine's 6DOF FMPE validation timings were recorded while other jobs shared the GPU.

5DOF Validation Results

200 validation poses × 1000 model samples per pose. Err_post and Err_resim are arithmetic means. Rows are sorted by increasing Err_post within this table only.

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	FMPE Midpoint 32	0.00307	0.00054	4.08488 / 11.87245	0.63029 / 2.00929	0.06183	0.97465	19.98471
2	FMPE Midpoint 16	0.00307	0.00057	5.30590 / 13.87356	0.73487 / 1.93526	0.06632	0.97453	9.96921
3	FMPE Midpoint 8	0.00310	0.00119	7.19194 / 19.59317	1.00860 / 2.67192	0.06836	0.97401	5.01582
4	FMPE Euler 64	0.00315	0.00048	3.12673 / 9.70195	0.47725 / 1.65997	0.05553	0.97328	20.07352
5	FMPE Euler 32	0.00332	0.00062	3.74304 / 11.03925	0.53678 / 1.67459	0.05112	0.97133	10.00179
6	CFMPE Midpoint 32	0.00373	0.00115	4.84918 / 18.56659	0.73137 / 3.00235	0.06675	0.97154	58.63389
7	CFMPE Midpoint 16	0.00373	0.00105	5.08302 / 16.73645	0.71993 / 2.66403	0.06983	0.97154	29.08537
8	CFMPE Midpoint 8	0.00375	0.00149	6.80326 / 20.40755	0.94098 / 3.01270	0.07448	0.97099	14.20444
9	CFMPE Euler 64	0.00379	0.00104	3.84538 / 15.74501	0.57449 / 2.62455	0.06053	0.96992	58.74780
10	CFMPE Euler 32	0.00392	0.00122	4.08502 / 14.95542	0.58070 / 2.42865	0.05622	0.96774	29.29047
11	FFF Euclidean	0.01602	0.25891	32.23237 / 264.42786	5.06740 / 80.38028	0.31029	0.84177	0.17750
12	M-FFF torus wrapped	0.02281	0.58592	54.95619 / 243.05582	16.76678 / 116.39543	0.28241	0.78820	0.18225
13	FIF	0.03014	0.44806	87.21315 / 396.62167	13.63169 / 99.80828	0.34092	0.66914	0.59784

Local FIF/CFMPE: GTX 1650 Ti. Emine: RTX 3060 Laptop. All FR3 training seeds are 0. Time measures model sampling only; timings on different hardware do not support a direct speed ranking.

Reference note: the local validation cache hashes differ from those reported by Emine. The target poses and sample counts are shared, but the reference samples are not identical.

Results are rounded to five decimal places. A displayed 0.00000 can represent a small nonzero value.

For 5DOF, Rot denotes the flange-axis angular error rather than full rotation error. The coverage radius is 0.50000 rad for 5DOF and 0.10000 rad for 6DOF.

FR3 Test Results

1000 test poses × 1000 samples per pose for each task. Err_post and Err_resim are arithmetic means. Rows are sorted by increasing Err_post within each task.

6DOF Test Results

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	Oracle	-0.00008	0.00000	0.00003 / 0.00007	0.00001 / 0.00001	0.00000	0.99463	—
2	FMPE Euler 64	0.02207	0.00493	4.57064 / 14.32152	1.10723 / 3.74252	0.08406	0.96660	100.12722
3	CFMPE Euler 64	0.02741	0.00698	5.93151 / 23.67971	1.29697 / 5.37915	0.09116	0.93327	310.45147
4	M-FFF torus wrapped	0.11682	2.24530	106.47573 / 275.44460	63.53410 / 166.33608	0.30372	0.20183	0.81090
5	FFF Euclidean	0.13524	4.02319	47.90350 / 272.84695	100.51644 / 172.16756	0.30708	0.02550	0.83100
6	FIF	0.15562	1.97056	193.13302 / 858.10392	45.63359 / 162.11481	0.07408	0.00157	3.00913

5DOF Test Results

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	Oracle	0.00000	0.00000	0.00003 / 0.00007	0.00000 / 0.00001	0.00000	0.98207	—
2	FMPE Euler 64	0.00319	0.00051	3.29702 / 10.25342	0.48818 / 1.68153	0.06003	0.97525	100.88697
3	CFMPE Euler 64	0.00385	0.00121	4.03223 / 15.87672	0.59496 / 2.61333	0.06258	0.97118	297.89331
4	FFF Euclidean	0.01674	0.24952	32.54252 / 264.31990	5.06319 / 78.16233	0.32096	0.84840	0.82629
5	M-FFF torus wrapped	0.02379	0.57634	53.06153 / 241.89255	16.25344 / 116.74309	0.29335	0.79402	0.80116
6	FIF	0.03207	0.45652	90.33020 / 419.70598	14.45037 / 99.60692	0.34753	0.67236	2.83667

FIF/CFMPE: local GTX 1650 Ti. FMPE/FFF/M-FFF: Emine's RTX 3060 Laptop. Timings are not directly comparable across hardware. Euler 64 was selected on validation for both CFMPE tasks; neither test required non-finite redraws.

The local test cache hashes differ from Emine's. Target poses and sample counts are shared, but reference samples are not identical. Oracle uses the original reference records.

Oracle is an independent reference draw estimating the error floor. Its timing is omitted because it was measured on another machine. Small negative Err_post values are possible with an unbiased MMD^2 estimator.

For 5DOF, Rot is the flange-axis angular error. Lim Viol and Cov are fractions between 0.00000 and 1.00000. A displayed 0.00000 may be a rounded nonzero value. Validation and test results are ranked separately.

Umut Planar Kinematics Results

Umut's recorded INN, cINN and iLNN evaluations. Each row is a separate training seed or variant. Err_post and Err_resim are means over 1000 conditions, with 1000 model samples per condition.

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	cINN s1 [U]	0.00306	0.00002	—	—	—	—	333.63694
2	INN ML s2 [U]	0.00338	0.00011	—	—	—	—	305.42124
3	cINN s0 [U]	0.00352	0.00001	—	—	—	—	324.65757
4	cINN s2 [U]	0.00354	0.00001	—	—	—	—	332.90267
5	INN ML s1 [U]	0.00362	0.00025	—	—	—	—	327.64260
6	INN ML s0 [U]	0.00366	0.00023	—	—	—	—	299.11331
7	iLNN ML s0 mix [U]	0.00438	0.00087	—	—	—	—	292.95163
8	iLNN ML s0 h96b16all [U]	0.00452	0.00030	—	—	—	—	84.03100
9	iLNN ML s0 [U]	0.00489	0.00013	—	—	—	—	308.76719
10	iLNN ML s0 sy02 [U]	0.00634	0.02762	—	—	—	—	349.69846
11	INN L2+MMD s1 [U]	0.01086	0.00797	—	—	—	—	235.07680
12	INN L2+MMD s2 [U]	0.01481	0.05426	—	—	—	—	233.39307
13	INN L2+MMD s0 [U]	0.02002	599845.01967	—	—	—	—	215.99665
14	iLNN MMD s0 [U]	0.02289	0.00275	—	—	—	—	360.22411
15	iLNN MMD s1 [U]	0.02353	0.00381	—	—	—	—	328.10975
16	iLNN original s0 [U]	0.02577	0.00281	—	—	—	—	377.15970

[U] Umut. These are different tasks from FR3 5DOF/6DOF and should not be ranked against the FR3 results. The JSON records specify device=cpu and a minimum reference sample count of 444. Dashes indicate that the FR3-specific Pos, Rot, Lim Viol and Cov fields are unavailable.

h96b16all, mix and sy02 preserve the original run variant names. The large Err_resim for INN L2+MMD s0 is the raw mean from its source file; outliers have not been clipped.

Umut Ballistics Results

Umut's recorded INN, cINN and iLNN evaluations. Each row is a separate training seed or variant. Err_post and Err_resim are means over 1000 conditions, with 1000 model samples per condition.

#	Model	Err_post ↓	Err_resim ↓	Pos (mm) Med/P95 ↓	Rot (deg) Med/P95 ↓	Lim Viol ↓	Cov ↑	Time (s) ↓
1	cINN s2 [U]	0.00472	0.00088	—	—	—	—	300.52004
2	cINN s1 [U]	0.00697	0.00002	—	—	—	—	286.32592
3	iLNN ML s0 [U]	0.00711	0.00011	—	—	—	—	349.67572
4	INN L2+MMD s1 [U]	0.00864	0.00325	—	—	—	—	208.28334
5	iLNN original s0 [U]	0.00871	0.01294	—	—	—	—	325.56084
6	INN L2+MMD s0 [U]	0.00873	0.00408	—	—	—	—	180.81449
7	iLNN MMD s0 [U]	0.00884	0.01395	—	—	—	—	310.85998
8	INN L2+MMD s2 [U]	0.00947	0.02528	—	—	—	—	200.36210
9	cINN s0 [U]	0.01013	0.12261	—	—	—	—	277.39427
10	INN ML s2 [U]	0.01707	1.11093	—	—	—	—	330.49892
11	INN ML s0 [U]	0.01927	0.01829	—	—	—	—	308.63164
12	INN ML s1 [U]	0.02978	0.00930	—	—	—	—	303.52223

[U] Umut. These are different tasks from FR3 5DOF/6DOF and should not be ranked against the FR3 results. The JSON records specify device=cpu and a minimum reference sample count of 111. Dashes indicate that the FR3-specific Pos, Rot, Lim Viol and Cov fields are unavailable.

Samples with an undefined forward result are excluded from Err_resim. Failed sample fractions in this table: UB-01: 0.00001; UB-04: 0.00047; UB-05: 0.00051; UB-06: 0.00046; UB-07: 0.00046; UB-08: 0.00292; UB-09: 0.00022; UB-10: 0.00041; UB-11: 0.00008; UB-12: 0.00015.

Metrics and Comparison Conditions

Meaning of the Means

Err_post is the arithmetic mean across target conditions of the unbiased MMD² estimate between model and reference samples. Err_resim is ||f(q) − y||² averaged over samples and then over conditions. Values come from the JSON mean fields. Clipped means, medians and training losses are not substituted. Different solver settings or training seeds are not averaged together.

Table Columns

Pos and Rot show the median and 95th percentile across model samples. Rot is the geodesic rotation error for 6DOF and the flange-axis angular error for 5DOF. Lim Viol is the fraction of samples violating at least one joint limit. Cov is the mean fraction of reference samples with a model sample inside the coverage radius: 0.10000 rad for 6DOF and 0.50000 rad for 5DOF. Time is inference_seconds; it excludes reference generation and metric computation. Results use five decimal places, so small nonzero values can round to 0.00000. ↓ means lower is better; ↑ means higher is better.

Validation and Test Splits

FR3 validation uses 200 poses and test uses 1000 poses, with 1000 model samples and 1000 reference samples per pose. Umut's available evaluation records in this checkout cover planar kinematics and ballistics; there are no FR3 result records for those models. Different tasks and splits are shown in separate tables.

Methods and Timings

Five CFMPE solver settings were compared on validation; Euler 64 was selected for both test tasks. CFMPE was trained on a contracted joint-prior family with a 25% nominal-prior anchor and evaluated at the nominal full-joint-limit prior. FIF was trained on the fixed nominal prior. Local FIF/CFMPE timings use a GTX 1650 Ti; Emine's FR3 timings use an RTX 3060 Laptop. Her 6DOF FMPE validation runs shared the GPU with other work. Umut's records specify CPU execution without identifying the machine.

Reference File Consistency

Emine reports reusing the same four reference files for FMPE and FFF/M-FFF. The local FIF/CFMPE validation and test files were regenerated and their hashes differ from hers. Validation hashes are listed below. A controlled comparison using identical reference samples requires evaluating the methods against shared cache files.

gt_fr3_6d_n1000_s777_3297457ebd.npz
Local SHA256 7b0c776ac5cc2fd769882ef779aebdd892be8925aaa2e46413e9498e85d3876e
Emine SHA256 28a1c3cb09ac0a98cae41165ac7a991bba47eb861bd69ce3c205756b9dcefe3b
gt_fr3_5d_n1000_s777_068e186e5c.npz
Local SHA256 c79a62d9a9acaedd5283bbb4b086f7545c64ef94fb2aca09aaeb054df7dc72b2
Emine SHA256 90a6fa733143b6a1d8da5679857a327beca5bf08ccf8d80f7751bd0fc384182a

Verification

The Err_post and Err_resim means for all 67 result records were checked against the per-condition arrays in the accompanying .npz files. No new training, sampling or reference generation was performed. The following pages list the source file for every table row.

Source Result Files

All files are under metrics/results/. Each code identifies a table and row: V6/V5 validation, T6/T5 test, UK planar kinematics, UB ballistics.

V6-01 fr3_6d_fr3_6d_fmpe_s0_midpointx32_val.json
V6-02 fr3_6d_fr3_6d_fmpe_s0_eulerx64_val.json
V6-03 fr3_6d_fr3_6d_fmpe_s0_midpointx16_val.json
V6-04 fr3_6d_fr3_6d_fmpe_s0_eulerx32_val.json
V6-05 fr3_6d_fr3_6d_fmpe_s0_midpointx8_val.json
V6-06 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx32_val.json
V6-07 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx64_val.json
V6-08 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx16_val.json
V6-09 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx8_val.json
V6-10 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx32_val.json
V6-11 fr3_6d_fr3_6d_mfff_wrapped_b400_s0_val.json
V6-12 FFF validation result with observation noise 0.01000 (source JSON)
V6-13 fr3_6d_fr3_6d_fff_b400_s0_val.json
V6-14 fr3_6d_fr3_6d_fif_s0_val.json
V5-01 fr3_5d_fr3_5d_fmpe_s0_midpointx32_val.json
V5-02 fr3_5d_fr3_5d_fmpe_s0_midpointx16_val.json
V5-03 fr3_5d_fr3_5d_fmpe_s0_midpointx8_val.json
V5-04 fr3_5d_fr3_5d_fmpe_s0_eulerx64_val.json
V5-05 fr3_5d_fr3_5d_fmpe_s0_eulerx32_val.json
V5-06 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx32_val.json
V5-07 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx16_val.json
V5-08 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_midpointx8_val.json
V5-09 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx64_val.json
V5-10 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx32_val.json
V5-11 fr3_5d_fr3_5d_fff_b400_s0_val.json
V5-12 fr3_5d_fr3_5d_mfff_wrapped_b400_s0_val.json
V5-13 fr3_5d_fr3_5d_fif_s0_val.json
T6-01 fr3_6d_oracle.json
T6-02 fr3_6d_fr3_6d_fmpe_s0_eulerx64.json
T6-03 fr3_6d_fr3_6d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx64.json
T6-04 fr3_6d_fr3_6d_mfff_wrapped_b400_s0.json
T6-05 fr3_6d_fr3_6d_fff_b400_s0.json
T6-06 fr3_6d_fr3_6d_fif_s0.json
T5-01 fr3_5d_oracle.json

Source Result Files

All files are under metrics/results/. Each code identifies a table and row: V6/V5 validation, T6/T5 test, UK planar kinematics, UB ballistics.

T5-02 fr3_5d_fr3_5d_fmpe_s0_eulerx64.json
T5-03 fr3_5d_fr3_5d_cfmpe_s0_width1-1-1-1-1-1-1_eulerx64.json
T5-04 fr3_5d_fr3_5d_fff_b400_s0.json
T5-05 fr3_5d_fr3_5d_mfff_wrapped_b400_s0.json
T5-06 fr3_5d_fr3_5d_fif_s0.json
UK-01 kinematics_kin_cinn_s1.json
UK-02 kinematics_kin_inn_ml_s2.json
UK-03 kinematics_kin_cinn_s0.json
UK-04 kinematics_kin_cinn_s2.json
UK-05 kinematics_kin_inn_ml_s1.json
UK-06 kinematics_kin_inn_ml_s0.json
UK-07 kinematics_kin_ilnn_ml_s0_mix.json
UK-08 kinematics_kin_ilnn_ml_s0_h96b16all.json
UK-09 kinematics_kin_ilnn_ml_s0.json
UK-10 kinematics_kin_ilnn_ml_s0_sy02.json
UK-11 kinematics_kin_inn_l2mmd_s1.json
UK-12 kinematics_kin_inn_l2mmd_s2.json
UK-13 kinematics_kin_inn_l2mmd_s0.json
UK-14 kinematics_kin_ilnn_mmd_s0.json
UK-15 kinematics_kin_ilnn_mmd_s1.json
UK-16 kinematics_kin_ilnn_s0.json
UB-01 ballistics_bal_cinn_s2.json
UB-02 ballistics_bal_cinn_s1.json
UB-03 ballistics_bal_ilnn_ml_s0.json
UB-04 ballistics_bal_inn_l2mmd_s1.json
UB-05 ballistics_bal_ilnn_s0.json
UB-06 ballistics_bal_inn_l2mmd_s0.json
UB-07 ballistics_bal_ilnn_mmd_s0.json
UB-08 ballistics_bal_inn_l2mmd_s2.json
UB-09 ballistics_bal_cinn_s0.json
UB-10 ballistics_bal_inn_ml_s2.json
UB-11 ballistics_bal_inn_ml_s0.json
UB-12 ballistics_bal_inn_ml_s1.json

Method and Protocol Sources

metrics/evaluate.py; metrics/mmd.py; robot_ik/metrics.py; docs/robot_ik.md.
Emine FMPE: docs/reports/2026-11-11_fr3-fmpe.md, commit f443004.
Emine FFF/M-FFF: docs/reports/2026-11-11_fr3-fff-mfff.md, commit 705a965.
Umut ILNN variants: e21300a; INN/cINN and earlier ILNN records: metrics/results/.